Programming in C Lab

Practice Set-1

Topics Covered

- Taking input in C
- Printing output
- `float` data type
- Basic arithmetic operations
- Divisibility check
- Modulus (`%`) operator

Instructions

Write a C program for each of the following questions.

- Take the required input from the user.
- Display the output clearly.
- Use appropriate data types.
- Use the modulus (`%`) operator for divisibility checks.

Practice Questions

1. **Float Input and Output**
   Write a C program to input a floating-point number and display it.

2. **Sum of Two Float Numbers**
   Write a C program to input two floating-point numbers and display their sum.

3. **Arithmetic Operations**
   Write a C program to input two floating-point numbers and display their:

   - Sum

- Difference
- Product
- Quotient

4. **Area of a Circle**
   Write a C program to input the radius of a circle as a `float` and calculate its area.

$$Area = \pi r^2$$

5. **Average of Three Numbers**
   Write a C program to input three floating-point numbers and calculate their average.

$$Average = \frac{a + b + c}{3}$$

6. **Percentage Calculation**
   Write a C program to input marks obtained in five subjects as floating-point values and calculate the total and percentage.

7. **Divisibility by 2**
   Write a C program to input an integer and check whether it is divisible by 2 using the modulus (`%`) operator.

8. **Divisibility by 5**
   Write a C program to input an integer and check whether it is divisible by 5 using the modulus (`%`) operator.

9. **Divisibility by 3 and 7**
   Write a C program to input an integer and check whether it is divisible by both 3 and 7.

10. **Multiple Divisibility Check**
    Write a C program to input an integer and check whether it is divisible by:

    - 2
    - 3
    - 5
    - 7

Important Concept

For checking whether an integer `n` is divisible by another integer `x`, use:

```
n % x == 0
```

If the result is `0`, then `n` is divisible by `x`.

Example

```
if (n % 5 == 0) {
    printf("Divisible by 5");
}
else {
    printf("Not divisible by 5");
}
```

Note

The modulus (%) operator is generally used with integer operands in these basic divisibility problems. For floating-point calculations, use appropriate arithmetic operators.